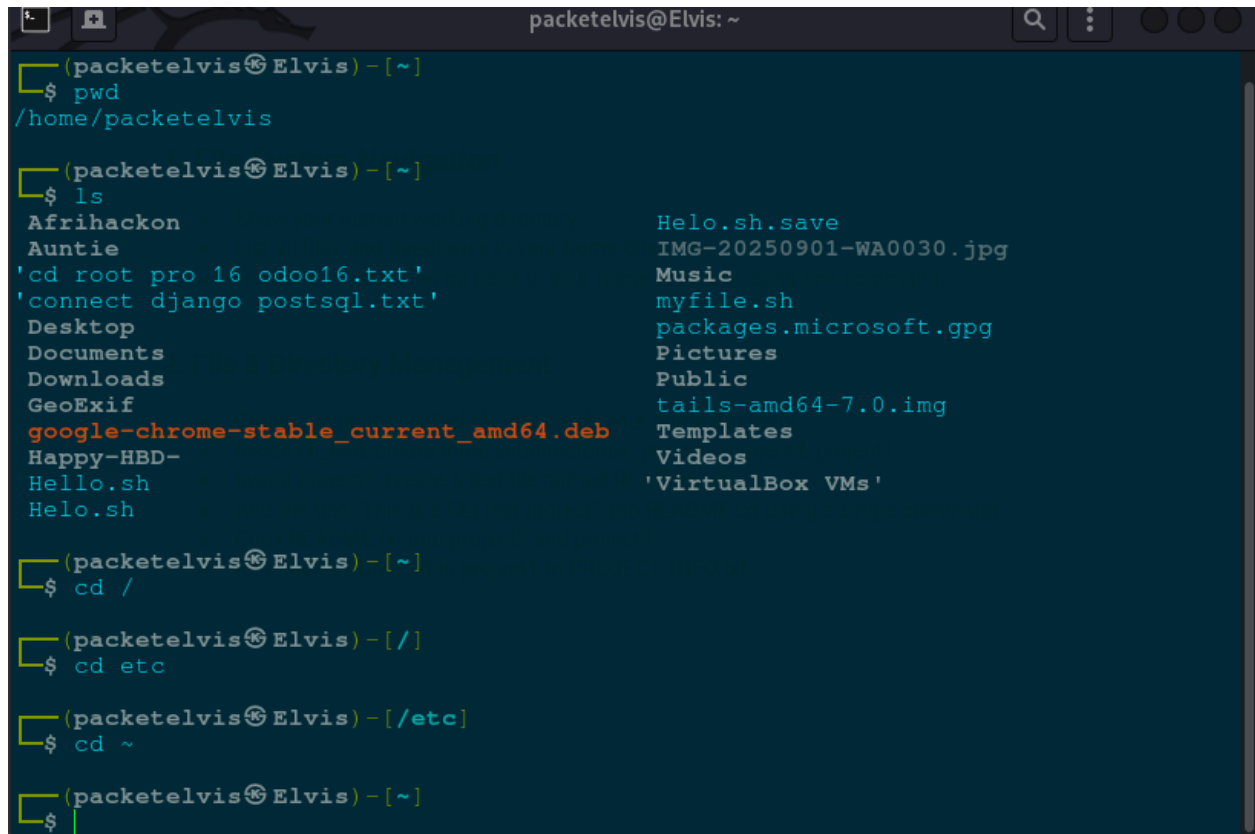


1. File System Navigation

- Show your current working directory.
- List all files and directories in your home directory.
- Navigate to /etc, then go back to your home directory using one command.

A terminal window titled 'packetelvis@Elvis: ~' showing a series of commands and their outputs. The user starts in their home directory (~). They run 'pwd' and get '/home/packetelvis'. Then they run 'ls' and see a list of files and directories including 'Afrihackon', 'Auntie', 'Desktop', 'Downloads', 'GeoExif', 'google-chrome-stable_current_amd64.deb', 'Happy-HBD-', 'Hello.sh', 'Helo.sh', 'Helo.sh.save', 'IMG-20250901-WA0030.jpg', 'Music', 'myfile.sh', 'packages.microsoft.gpg', 'Pictures', 'Public', 'tails-amd64-7.0.img', 'Templates', 'Videos', and 'VirtualBox VMs'. Next, they run 'cd /' to change to the root directory, then 'cd etc' to move to /etc, and finally 'cd ~' to return to their home directory. The prompt returns to '\$' in the home directory.

```
(packetelvis@Elvis) - [~]
$ pwd
/home/packetelvis

(packetelvis@Elvis) - [~]
$ ls
Afrihackon
Auntie
'cd root pro 16 odoo16.txt'
'connect django postgresql.txt'
Desktop
Downloads
GeoExif
google-chrome-stable_current_amd64.deb
Happy-HBD-
Hello.sh
Helo.sh
Helo.sh.save
IMG-20250901-WA0030.jpg
Music
myfile.sh
packages.microsoft.gpg
Pictures
Public
tails-amd64-7.0.img
Templates
Videos
'VirtualBox VMs'

(packetelvis@Elvis) - [~]
$ cd /

(packetelvis@Elvis) - [/]
$ cd etc

(packetelvis@Elvis) - [/etc]
$ cd ~

(packetelvis@Elvis) - [~]
$
```

pwd Get the full path of the current working directory.

ls lists all **files and directories** within the current or specified path.

cd /etc && cd ~ changes the directory to **/etc** and then returns to your **home directory** in one combined command.

2. File & Directory Management

- Create a directory called cli_test inside your home directory.
- Inside cli_test, create three subdirectories: project1, project2, project3.
- Inside project1, create a text file named README.txt.
- Add the text "This is a CLI test project" into README.txt using a single command.

- Copy README.txt into project2 and project3.
- Rename README.txt in project1 to PROJECT_INFO.txt.

```

packetelvis@Elvis: ~/Afrihackon/Linux/cli_test x root@Elvis: ~ x
└─(packetelvis@Elvis) - [~/Afrihackon/Linux]
$ mkdir cli_test

└─(packetelvis@Elvis) - [~/Afrihackon/Linux]
$ cd cli_test/\
>

└─(packetelvis@Elvis) - [~/Afrihackon/Linux/cli_test]
$ mkdir project1 project2 project3

└─(packetelvis@Elvis) - [~/Afrihackon/Linux/cli_test]
$ cd project1

└─(packetelvis@Elvis) - [~/Afrihackon/Linux/cli_test/project1]
$ nano README.txt

└─(packetelvis@Elvis) - [~/Afrihackon/Linux/cli_test/project1]
$ cd ..

└─(packetelvis@Elvis) - [~/Afrihackon/Linux/cli_test]
$ cp -p project1/README.txt project2/ project3/
cp: -r not specified; omitting directory 'project2/'

└─(packetelvis@Elvis) - [~/Afrihackon/Linux/cli_test]
$ cp -p project1/README.txt project2/

└─(packetelvis@Elvis) - [~/Afrihackon/Linux/cli_test]
$ cp -p project1/README.txt project3/

└─(packetelvis@Elvis) - [~/Afrihackon/Linux/cli_test]
$ mv project1/README.txt project1/PROJECT_INFO.txt

```

mkdir cli_test — Creates a new directory named `cli_test` inside the current working directory.

cd cli_test/ — Changes the current directory to `cli_test`.

mkdir project1 project2 project3 — Creates three subdirectories named `project1`, `project2`, and `project3`.

cd project1 — Moves into the `project1` directory.

nano README.txt — Opens a text editor to create and edit the file `README.txt`.

cp -p project1/README.txt project2/ project3/ — Copies the README.txt file from project1 to both project2 and project3.

mv project1/README.txt project1/PROJECT_INFO.txt — Renames README.txt in project1 to PROJECT_INFO.txt.

```
(packetelvis@Elvis) - [~/Afrihackon/Linux/cli_test]
$ cd project1

(packetelvis@Elvis) - [~/Afrihackon/Linux/cli_test/project1]
$ cat PROJECT_INFO.txt
This is a CLI test project

(packetelvis@Elvis) - [~/Afrihackon/Linux/cli_test/project1]
$
```

Cat Project_info.txt → to view what the file contains

3File & Directory Cleanup

- Move PROJECT_INFO.txt from project1 to the home directory.
- Delete the entire cli_test directory and its contents.

```
(packetelvis@Elvis) - [~/Afrihackon/Linux/cli_test/project1]
$ mv PROJECT_INFO.txt ~/

(packetelvis@Elvis) - [~/Afrihackon/Linux/cli_test/project1]
$ cd ..

(packetelvis@Elvis) - [~/Afrihackon/Linux/cli_test]
$ rm -r ~/Afrihackon/Linux/cli_test

(packetelvis@Elvis) - [~/Afrihackon/Linux/cli_test]
$ cd ..

(packetelvis@Elvis) - [~/Afrihackon/Linux]
$ ls

(packetelvis@Elvis) - [~/Afrihackon/Linux]
$
```

mv PROJECT_INFO.txt ~/ — Moves the file PROJECT_INFO.txt from the current directory (project1) to the user's home directory.

cd .. — Moves one level up in the directory hierarchy (from project1 to cli_test).

rm -r ~/Afrihackon/Linux/cli_test – Recursively deletes the cli_test directory and all its contents.

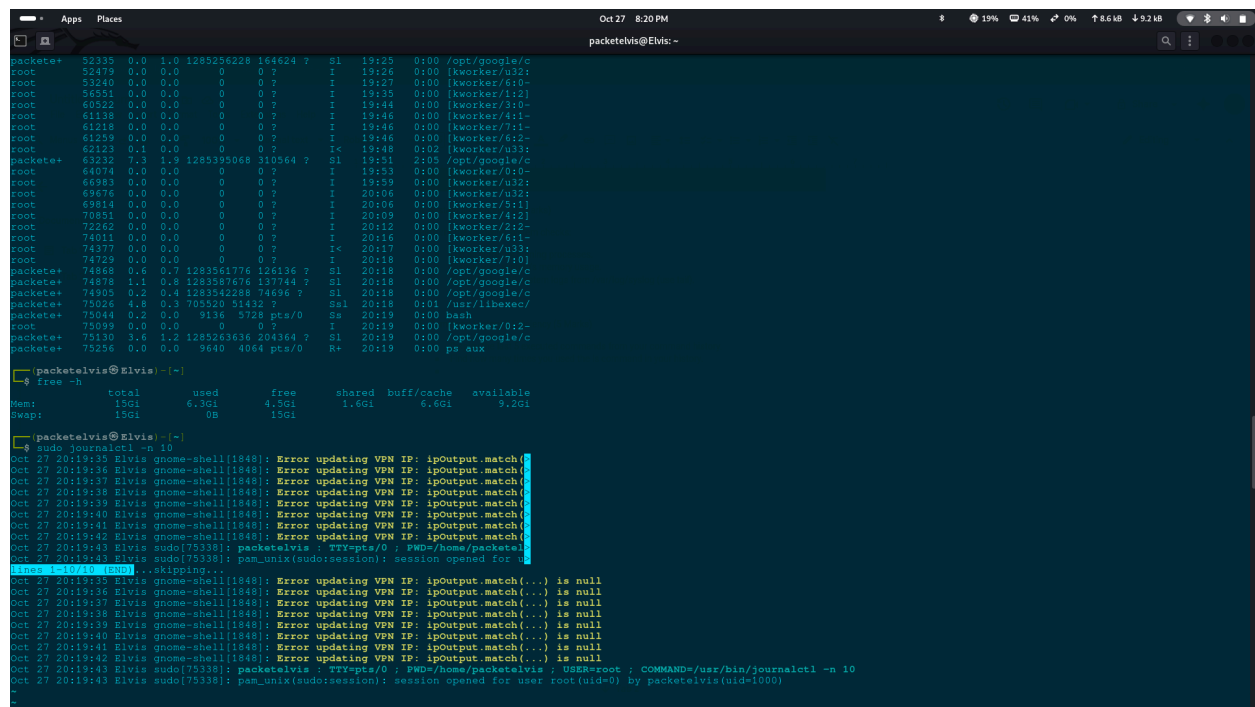
cd .. – Moves back to the parent directory (Linux).

ls – Lists the remaining files and directories, confirming that cli_test has been successfully deleted.

4. System Monitoring

Perform the following system checks:

- List all currently running processes.
- Display your system's memory usage.
- Show the last 10 system logs from /var/log/syslog (use tail).



```
packetelvis@Elvis:~$ ps aux
packetelvis 52335 0.0 1.0 1285256228 164624 ? S1 19:25 0:00 /opt/google/c
root 52479 0.0 0.0 0 0 ? I 19:26 0:00 [kworker/0:2]
root 53240 0.0 0.0 0 0 ? I 19:27 0:00 [kworker/6:0]
root 56551 0.0 0.0 0 0 ? I 19:35 0:00 [kworker/1:2]
root 60522 0.0 0.0 0 0 ? I 19:44 0:00 [kworker/3:0]
root 61138 0.0 0.0 0 0 ? I 19:46 0:00 [kworker/4:1]
root 61218 0.0 0.0 0 0 ? I 19:46 0:00 [kworker/7:1]
root 61259 0.0 0.0 0 0 ? I 19:46 0:00 [kworker/6:2]
root 62162 0.1 0.0 0 0 ? I 19:46 0:02 [kworker/u3:1]
packetelvis 63232 7.3 1.9 1285395068 310564 ? S1 19:51 2:05 /opt/google/c
root 64074 0.0 0.0 0 0 ? I 19:53 0:00 [kworker/0:0]
root 64983 0.0 0.0 0 0 ? I 19:59 0:00 [kworker/0:2]
root 69676 0.0 0.0 0 0 ? I 20:06 0:00 [kworker/u3:2]
root 69814 0.0 0.0 0 0 ? I 20:06 0:00 [kworker/5:1]
root 70851 0.0 0.0 0 0 ? I 20:09 0:00 [kworker/4:2]
root 72262 0.0 0.0 0 0 ? I 20:12 0:00 [kworker/2:2]
root 74011 0.0 0.0 0 0 ? I 20:16 0:00 [kworker/6:1]
root 74177 0.0 0.0 0 0 ? I 20:17 0:00 [kworker/u3:3]
root 74729 0.0 0.0 0 0 ? I 20:18 0:00 [kworker/7:0]
packetelvis 74868 0.6 0.7 1283961776 126136 ? S1 20:18 0:00 /opt/google/c
packetelvis 74878 1.1 0.8 128387676 137744 ? S1 20:18 0:00 /opt/google/c
packetelvis 74905 0.2 0.3 1283542486 74656 ? S1 20:18 0:00 /opt/google/c
packetelvis 75026 4.8 0.3 705520 51432 ? Ssl 20:18 0:01 /usr/libexec/
packetelvis 75044 0.2 0.0 9136 5728 pts/0 Ss 20:19 0:00 bash
root 75099 0.0 0.0 0 0 ? I 20:19 0:00 [kworker/0:2]
packetelvis 75130 3.6 1.2 1285263636 204364 ? S1 20:19 0:00 /opt/google/c
packetelvis 75256 0.0 0.0 9640 4064 pts/0 R+ 20:19 0:00 ps aux

packetelvis@Elvis:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           15Gi        6.3Gi        4.5Gi        1.6Gi        6.6Gi        9.2Gi
Swap:           15Gi          0B          15Gi

packetelvis@Elvis:~$ sudo journalctl -n 10
Oct 27 20:19:35 Elvis gnome-shell[1848]: Error updating VPN IP: ipOutput.match(...) is null
Oct 27 20:19:36 Elvis gnome-shell[1848]: Error updating VPN IP: ipOutput.match(...) is null
Oct 27 20:19:37 Elvis gnome-shell[1848]: Error updating VPN IP: ipOutput.match(...) is null
Oct 27 20:19:38 Elvis gnome-shell[1848]: Error updating VPN IP: ipOutput.match(...) is null
Oct 27 20:19:39 Elvis gnome-shell[1848]: Error updating VPN IP: ipOutput.match(...) is null
Oct 27 20:19:40 Elvis gnome-shell[1848]: Error updating VPN IP: ipOutput.match(...) is null
Oct 27 20:19:41 Elvis gnome-shell[1848]: Error updating VPN IP: ipOutput.match(...) is null
Oct 27 20:19:42 Elvis gnome-shell[1848]: Error updating VPN IP: ipOutput.match(...) is null
Oct 27 20:19:43 Elvis sudo[75338]: packetelvis : TTY=pts/0 ; PWD=/home/packetelvis ; USER=root ; COMMAND=/usr/bin/journalctl -n 10
Oct 27 20:19:43 Elvis sudo[75338]: pam_unix(sudo:session): session opened for user root(uid=0) by packetelvis(uid=1000)
```

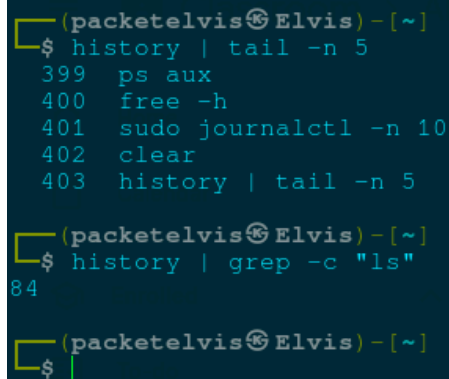
ps aux → Displays a detailed list of all running processes, including the user, CPU/memory usage, and command being executed.

free -h → Shows the total, used, and available memory (RAM and swap) in a human-readable format (**-h** adds units like MB/GB).

sudo journalctl -n 10 → Prints the last 10 lines from the system log file to quickly view recent system messages or errors.

5. Command History & Efficiency

- Display the last 5 executed commands from your command history.
- Find how many times you used the **ls** command in your history.



```
(packetelvis@Elvis) ~  
$ history | tail -n 5  
399 ps aux  
400 free -h  
401 sudo journalctl -n 10  
402 clear  
403 history | tail -n 5  
  
(packetelvis@Elvis) ~  
$ history | grep -c "ls"  
84  
  
(packetelvis@Elvis) ~  
$
```

history | tail -n 5 → Shows the last 5 commands from your shell history by taking the output of history and passing it to tail to display only the last few lines.

history | grep -c "ls" → Searches your command history for occurrences of the word **ls** and counts (**-c**) how many times it appears.

Research from **Linux Notes for Professionals**